

Statement of Work Details

Project ID	UU1231
Project Name	RMU – AI Agent Discovery, Definition, and Development
Company	Rocky Mountain University
Sponsor Contact Name	Dan Harmuth
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Faculty Advisor Name	David Agogo
Faculty Advisor Email	david.agogo@eccles.utah.edu
Project Completion Semester	Summer (July)
Year	2026
Will all team members be taking IS 6803-001 together?	Yes
Student Learning Goals	1. Apply data pipeline and transformation techniques using a Data Warehouse environment to process student activity data. 2. Design and implement rule-based business logic for determining student participation and attendance status. 3. Develop skills in workflow automation using tools such as SharePoint and Microsoft Power Automate. 4. Build system integration capabilities across Canvas data sources, Data Warehouse tables, and downstream applications. 5. Practice system validation and testing, including comparing automated outputs against manual processes. 6. Strengthen team collaboration and project management skills, including milestone tracking and stakeholder communication. 7. Gain domain knowledge of academic attendance policies and registrar workflows.
Project Deliverables with Requirements	Deliverables: 1. Project Governance: Statement of Work (SOW) & Business Rules/Rationale. • Finalized SOW aligned with automation-based workflow

- Clearly defined attendance business rules, including:
 - Adjusted participation date logic
 - Engagement classification criteria
 - 1-week inactivity threshold
- Documentation of assumptions, constraints, and attendance policy alignment

2. Workflow Analysis: Current state Process Maps & Repository.

- Current-state process maps:
 - Manual attendance review workflow
 - Canvas + Excel validation process
- Future-state process maps:
 - Data Warehouse + automation + SharePoint + approval workflow
- Repository of:
 - Sample datasets (Canvas / Data Warehouse structure)
 - Data fields and definitions
 - Exception cases and edge scenarios

3. Design and Prototype: Conceptual Design & UI Mockups.

- Conceptual system design reflecting:
 - Data flow from Data Warehouse → SharePoint → Power Automate
- 8+ UI mockups/wireframes, including:
 - SharePoint list views (flagged students, approvals).
 - Approval interface (status updates, filtering, views).
 - Optional dashboard views (risk status, weekly summary).

4. System Architecture: Architecture Diagram & Data Flow.

- Architecture diagram (Visio or equivalent) showing:
 - Data Warehouse (source of truth)
 - Gold table structure
 - Automation script/agent layer
 - SharePoint integration
 - Power Automate workflows
- Data flow documentation, including:
 - Data extraction (SQL / API endpoint)
 - Participation inclusion criteria
 - End-to-end workflow from data → flagging → approval

5. Technical Solution: Automation Pipeline & Power Platform Assets.

- The primary interface is the SharePoint list
- Implemented data processing pipeline, including:
 - Gold table with:
 - Student ID, Course ID
 - Last participation (original & adjusted)
 - Participation flag (1/0)
- Participation logic implementation:
 - Last viewed / latest activity-based calculation
- Script/automation component to:
 - Retrieve processed data from the Data Warehouse
 - Populate the SharePoint list
- Power Platform assets:
 - SharePoint list (structured for review and approval)

6. Quality Assurance: Testing & Validation Artifacts.

- Validation phase implementation:
 - Validation Mode Output:
 - Full Dataset for auditing
- Validation report comparing:
 - Manual attendance decisions vs automated output
- Test artifacts:
 - Test cases (logic validation, workflow testing)
 - Results and identified discrepancies
 - Iterative improvements to logic
- Evidence of:
 - Accuracy of adjusted participation date
 - Correct flagging behavior

7. Handover & Training: Documentation & Enablement.

- User guide for Registrar staff:
 - Reviewing flagged students
 - Slide deck or walkthrough video
- Configuration guide:
 - Updating thresholds and rules

Requirements

Functional Requirements:

1 - Automated Attendance Processing

- Automatically retrieve student activity data from the Data Warehouse (via SQL query or API endpoint) instead of relying solely on Canvas exports
- Display the most recent participations in the gold table
- Execute weekly batch processing with optional on-demand processing
- Generate a processed dataset (gold table) including:
 - Student ID, Course ID
 - Most recent participation date
- Reduce or eliminate manual report review by automating data preparation and filtering steps

2 - Engagement Classification

- Differentiate between:
 - Canvas-recorded activity (logins, page views)
 - RMU-defined engagement (submissions, discussions, quizzes, meaningful content interaction)
- Apply refined participation logic:
 - Use "last viewed" or the latest meaningful interaction date instead of simple counts
 - Exclude non-participatory activities:
 - Course Home
 - Course Grades
 - Course People
 - Exclude irrelevant downloads (e.g., static/logo files)
 - Exclude activity before the semester start date
- Apply RMU attendance policy thresholds:
 - Flag students with no participation within the last 7 days
 - Re-flag students with no participation within the last 14 days

3 - Rule-Based Escalation Workflow

- Flag: Identify students for 1st or 2nd warnings based on attendance criteria (per course level)

4 - Validation

- System supports an initial validation phase:
 - Display flagged students
 - Compare:
 - Original participation date
 - Adjusted participation date
- Enable verification of system accuracy against the manual process
- Transition to production mode after validation:
 - Only flagged students are shown

5 - Data Output & Visibility (Dashboard)

- Centralized SharePoint list displaying:
 - Student name / ID
 - Course and section
 - Last participation (original vs adjusted).
 - Timestamp
 - (More required to be included).
- Support filtered views:
 - Weekly flagged students
 - Courses

Non-Functional Requirements:

1 - Security & Compliance

- Fully FERPA-compliant
- Use anonymized/test data during pilot
- Production data restricted to the RMU Microsoft 365 tenant
- Role-based access control and audit logging

2 - Usability

- Simple, intuitive interface via SharePoint
- Minimal training required for Registrar staff
- Clear visibility into flagged cases and actions

3 - Maintainability

- Built using Microsoft Power Platform:
 - Data Warehouse (SQL)
 - SharePoint
 - Power Automate
- Business rules configurable without major code changes
- Modular design for logic updates

4 - Scalability

- Support datasets of ~5,000+ records per run
- Handle multiple courses and programs simultaneously
- Expandable to additional departments

5 - Integration

- Data Warehouse (primary source of truth)
- SharePoint (review interface)
- Power Automate (workflow automation)
- Optional:
 - Canvas API (supplemental data)
 - SIS integration (future enhancement)

6 - Reliability & Monitoring

- Weekly batch execution with consistent output
- Logging of:
 - Data processing runs
- Error handling and alerts for failures
- Repeatable and stable workflow execution

7. Pilot Requirements

- Fully functional in a test environment
- Use anonymized or synthetic data
- Test end-to-end workflow:
 - Data → Flagging → Display

8. Documentation & Handover

- Technical maintenance guide
- Architecture diagram
- Training materials (videos or live demo)

Project Milestones with Dates

Draft SOW – March 8

Final SOW – April 26

Business Requirements
Document – *TBD*

Pilot Testing with
Registrar – *TBD*

Implementation Plan – *TBD*

Final Presentation – *TBD*

Project Risks and Assumptions

Risks

- Delays in access to Canvas API or data

- Integration challenges between Canvas and SIS
- Data quality or inconsistency in historical records
- User adoption challenges during pilot

Assumptions

- Access to the Data Warehouse and Canvas activity data will be provided.
- Required data fields (e.g., activity logs, timestamps) are available or can be derived.
- Registrar staff will be available for:
 - Requirement clarification
 - Validation and testing
- SharePoint and Power Automate environments are accessible and functional.
- Weekly processing schedule (Friday) aligns with Registrar operations.
- Manual approval will remain part of the workflow (system does not fully automate decisions).
- Email system integration is supported within Microsoft ecosystem.
- The system will operate within a controlled pilot environment before full deployment.

The cadence of our group meetings with our sponsor will be to have live meetings:

Every other week

On the weeks that we don't have a live meeting with our sponsor, we will communicate/update them via:

Teams

Attach a PDF of all members of your team, sponsor, and advisor approving the details of the SOW. Email is acceptable

If you selected "Other," please specify what avenue you will be using. If not then put N/A.

N/A

Project Manager Name

Sooyoung Lee

Student uIDs

U0759225
u1320679
U1551290

Team Member #2

Paul Arnold

Team Member #3

Robert Bost

Team Member #4

Team Member #5

Team Member #6

DRAFT: I, the project manager, certify that all team members have viewed the draft SOW.

Yes

FINAL: I, the project manager, certify that all team members have approved the details on the SOW.

Yes

FINAL: I, the project manager, certify that our Faculty Advisor AND Sponsor have approved the details on the SOW.

Yes

Project Manager Signature

SL

Team Member #2 Signature

Team Member #3 Signature

Team Member #4 Signature

Team Member #5 Signature

Team Member #6 Signature

Project Information UU1231 - RMU – AI Agent Discovery, Definition, and Development - Rocky Mountain University

Date 05/15/2026 12:00pm

SOW Complete No